

Dogecoin Price Prediction using Random Forest Regression

Research Paper Summary

Abstract

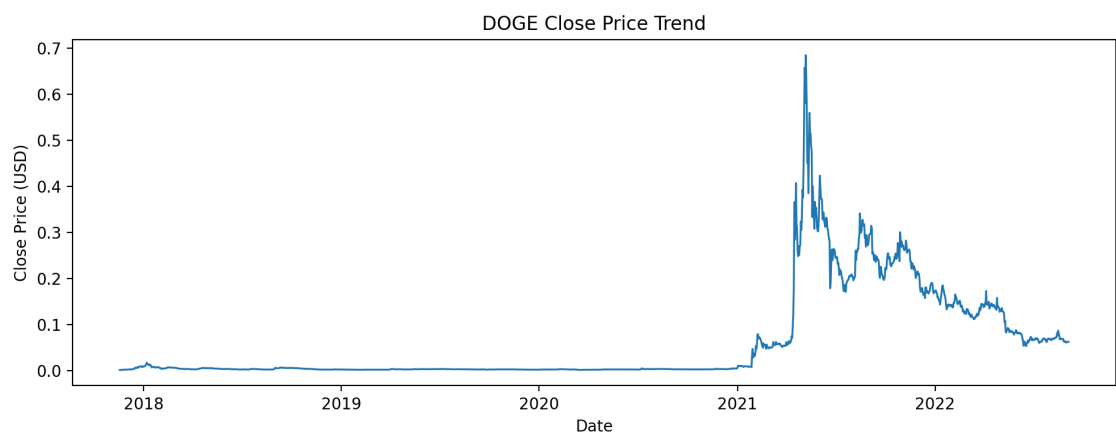
This paper investigates a machine learning approach for forecasting DOGE-USD close prices. The framework combines market inputs with technical indicators and evaluates a Random Forest Regressor using holdout testing metrics.

1. Problem Defined

Predict the close price of Dogecoin from open/high/low/volume values and engineered signals (MA5, MA10, Daily Return, Volatility).

2. Dataset Used

DOGE-USD.csv with 1761 raw rows and 1751 processed rows; date window 2017-11-18 to 2022-09-04; split 1400/351.



3. Model Trained

RandomForestRegressor configured with 100 estimators, max depth 15, min split 5, min leaf 2.

4. Accuracy Shown

Metric	Value
MSE	0.00001884
RMSE	0.00434057
MAE	0.00142236
R2	0.997776

5. Output Visible

Sample	Actual	Predicted
1	0.005545	0.005598

2	0.002346	0.002346
3	0.002434	0.002412
4	0.002083	0.002076
5	0.200887	0.203924

